

Name: _____

These questions assess your understanding of the material from Chapters 13, 14, 15 of OpenStax College Physics. Please write your name on the sheets of paper that you use and show all your work.

1. In a particular year, the main span of San Francisco's Golden Gate Bridge was 1272-m long on the coldest day. The bridge was exposed to temperatures ranging from -5°C to 42°C . What is the change in its length between these temperatures? Assume that the bridge is made entirely of steel so that its coefficient of linear expansion is $12 \times 10^{-6} \text{ } 1/^{\circ}\text{C}$.

ANSWER: _____

2. John's 80-L (21.1-gal) steel gasoline tank is full of gas. Both the tank and the gasoline have an initial temperature of 6.1°C . How much gasoline has leaked out due to thermal expansion by the time both the tank and the gasoline have warmed to 36.6°C ? The coefficients of volume expansion for steel and gasoline are respectively $35 \times 10^{-6} \text{ } 1/^{\circ}\text{C}$ and $950 \times 10^{-6} \text{ } 1/^{\circ}\text{C}$.

ANSWER: _____

3. Julia wants to use her pressure cooker to cook food in liquid water at a temperature of 160°C . To determine the pressure required inside the pressure cooker for water to boil at this temperature, she can use the Clausius-Clapeyron equation for water: $P(T) = (1 \text{ kPa}) \cdot \exp \left[(4750 \text{ K}) \left(\frac{1}{273.15 \text{ K}} - \frac{1}{T} \right) \right]$ In this equation, T is the temperature in Kelvin and P is the pressure in kPa. Given that the lid of the pressure cooker is a disk with a diameter of 37 cm, what is the force that the lid must be able to withstand at the required pressure?

ANSWER: _____

4. What is the average kinetic energy of a gas molecule at 133°C ?

ANSWER: _____

5. Patricia's 60-L (15.9-gal) steel gasoline tank is full of gas. Both the tank and the gasoline have an initial temperature of 11.9°C . After both the tank and the gasoline warm to 36.6°C , what is the pressure in the gasoline tank if the gasoline cannot spill out? The coefficients of volume expansion for steel and gasoline are respectively $35 \times 10^{-6} \text{ } 1/^{\circ}\text{C}$ and $950 \times 10^{-6} \text{ } 1/^{\circ}\text{C}$. Assume the bulk modulus B for gasoline is $1.00 \times 10^9 \text{ N/m}^2$.

ANSWER: _____

6. Brian puts a 0.69-kg aluminum pot on the stove and fills it with 0.8 liters of water. Both the pot and the water are initially at 13.6°C . How much heat is required to bring them both to 63.4°C ? The specific heat values of water and aluminum are $4186 \text{ J/kg}^{\circ}\text{C}$ and $900 \text{ J/kg}^{\circ}\text{C}$ respectively.

ANSWER: _____

7. Jason is producing heat at a rate of about 124 W while doing light exercise. At what rate must water evaporate from his body to get rid of the produced energy? The latent heat of vaporization of water is 2430 kJ/kg at body temperature, 37°C .

ANSWER: _____

8. Matthew is a design engineer for a lightbulb company. He manufactures a gas-filled incandescent light bulb so that the gas inside the bulb is at atmospheric pressure when the bulb has a temperature of 7°C . What is the pressure in the light bulb when it rises to 250°C ? The coefficient of volume expansion for the glass light bulb is $35 \times 10^{-6} \text{ } 1/^{\circ}\text{C}$.

ANSWER: _____

9. Calculate the work output of a Carnot engine operating between temperatures of 579 K and 289 K for 1400 J of heat transfer to the engine.

ANSWER: _____

10. Find the increase in entropy as 5.7 kg of ice, originally at 0.9°C , melts to form water at 0.9°C at a particular pressure. The latent heat of fusion for water is 334 kJ/kg .

ANSWER: _____

11. Spontaneous heat transfer from hot to cold is an irreversible process. Calculate the total change in entropy if 5270 J of heat transfer occurs from a hot reservoir at 537 K to a cold reservoir at 286 K, assuming that there is no temperature change in either reservoir.

ANSWER: _____

12. Benjamin inflates his bicycle tire to an absolute pressure of 0.49 MPa (a gauge pressure of about 56 psi) at a temperature of 7.4°C . What is the tire's gauge pressure after the temperature has risen to 30.7°C ? Assume there are no leaks and the volume of the tire does not change.

ANSWER: _____

13. Grace drives a 5900-kg truck and descends a 19-m-tall hill at a constant speed. What is the final temperature of 110 kg of brake material if its temperature is 9°C at the top of the hill? Assume the specific heat of the brake material is $830\text{ J/kg}^{\circ}\text{C}$. Assume the brakes do all the work to maintain the speed.

ANSWER: _____

14. Suppose that there is heat transfer of 49 J to a system, while the system does 43 J of work. Later, there is heat transfer of 33 J out of the system while 56 J of work is done on the system. What is the net change in internal energy of the system?

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